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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2022 (New Course)

Time: 3 hours

Code: 103402/110402 *Subject: ELECTRICAL MACHINES-II* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[2x7=14]**

(a) The pitch factor, in rotating electrical machinery, is defined as the ratio of resultant emf of a
- (i) full-pitched coil to that of a chorded coil
- (ii) full-pitched coil to the phase emf
- (iii) chorded coil to the phase emf
- (iv) chorded coil to that of a full-pitched coil

(b) For a 3 phase uniformly distributed narrow-spread winding, and the distribution factor is
- (i) 0.9
- (ii) 0.955
- (iii) 0.827
- (iv) 0.97

(c) MMF is related to
- (i) ampere turns
- (ii) voltage turns
- (iii) ampere hours
- (iv) voltage hours
- (v) None of the above

(d) When a single-phase winding is excited by an alternating current, then which of the following statements is correct?
- (i) It produces one stationary MMF wave
- (ii) Produced MMF wave pulsates along its magnetic axis
- (iii) It does not produce two counters rotating MMF wave
- (iv) All of the above

(e) Three-phase voltages are applied to the three windings of an electrical machine. If any two supply terminals are interchanged, then the rotating MMF wave
- (i) direction reverses, amplitude alters
- (ii) direction reverses, amplitude unaltered
- (iii) direction remains same, amplitude alters
- (iv) direction remains same, amplitude unaltered

(f) An induction motor can be said analogous to
- (i) transformer
- (ii) synchronous motor
- (iii) universal motor
- (iv) stepper motor

(g) No-load current in induction motor is 10%-20% of full-load current and the no-load current of transformer is 2%-6%.
- (i) True
- (ii) False

(h) Mechanically air gaps in induction motor are kept very low to avoid
- (i) lower power factor
- (ii) lagging nature
- (iii) magnetizing current
- (iv) All of the above

(i) A shaded pole motor is one of the types of
- (i) DC shunt motor
- (ii) differentially compounded DC motor
- (iii) synchronous motor
- (iv) single-phase induction motor

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(j) Which of the following special motors has rotor movement in discrete step?

(i) Stepper motor
(ii) Reluctance motor
(iii) Servomotor
(iv) Hysteresis motor

Q.2 Solve both questions : **[10+4=14]**

(a) Compare full pitch coil and chorded coil in a rotating machine. Discuss MMF of a coil along with its distribution in the winding. Explain with the help of suitable neat and labelled diagram.

(b) Discuss the following terms: (i) MMF, flux per pole (ii) Pole pitch (iii) Cylindrical and salient rotor (iv) Winding distribution factor

Q.3 Solve both questions : **[6+8=14]**

(a) Differentiate concentrated winding and distributed winding, with neat and labelled diagram. Discuss, why the windings are distributed sinusoidally.

(b) Explain, how pulsating magnetic field is produced in 1-phase induction motor. Whether it supports self-starting or not? If supports, how? If not, what measures do we take to make it start?

Q.4 Solve this question : **[14]**

(a) Discuss the following with suitable diagrams: (i) 1-Phase windings shifted by 90$\circ$ (ii) 3-Phase windings shifted by 120$\circ$

(b) Provide a basic constructional detail of 3 phase induction motor. Explain the role of wound rotor and squirrel cage type rotor in an induction motor.

Q.5 Solve this question : **[14]**

What are the three different modes of operation of an induction motor? Draw typical torque-slip characteristic, along with fan load characteristic, explaining the steady-state operating point.

Q.6 Solve both questions : **[7+7=14]**

(a) Why is blocked rotor test performed on an induction motor? What is the information perceived from this test?

(b) Draw and explain the equivalent circuit diagram of an induction motor.

Q.7 Solve both questions : **[8+6=14]**

(a) Mathematically and graphically explain how rotor resistance affects starting torque of an induction motor.

(b) A 3-phase, 50 Hz induction motor has a full-load speed of 1440 r.p.m. For this motor, calculate (i) number of poles; (ii) full load slip and rotor frequency. (iii) speed of stator field with respect to 1. stator structure and 2. rotor structure; (iv) speed of rotor field with respect to 1. rotor structure, 2. stator structure and 3. stator field.

Q.8 Solve this question : **[8+6=14]**

A 3-phase, 4-pole, 1440 r.p.m., 50 Hz induction motor has star-connected rotor winding, having a resistance of 0.2 Ω per phase and a standstill leakage reactance of 1 Ω per phase. When the stator is energised at rated voltage and frequency, the rotor induced e.m.f. at standstill is 120 V per phase.

Calculate the rotor current, rotor power factor and torque both at starting and at full load, and compare these results.

If an external resistance of 1 Ω per phase is inserted in rotor circuit, calculate the rotor current, rotor power factor and torque at the time of starting.

Q.9 Solve both questions : **[7+7=14]**

(a) Discuss parallel operation of alternators for synchronization and load division.

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(b) Develop the power equation for salient pole synchronous machine with the help of two-reaction theory.

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